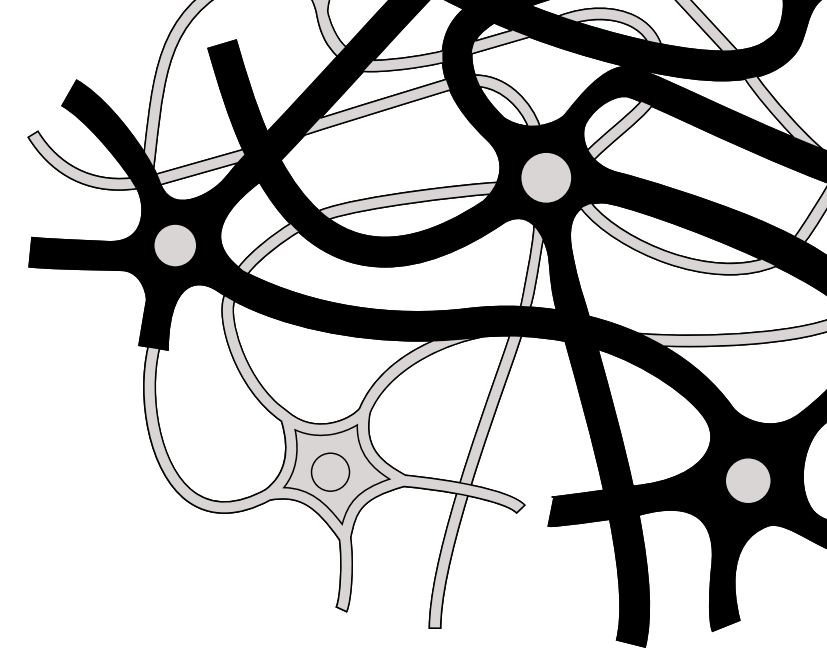
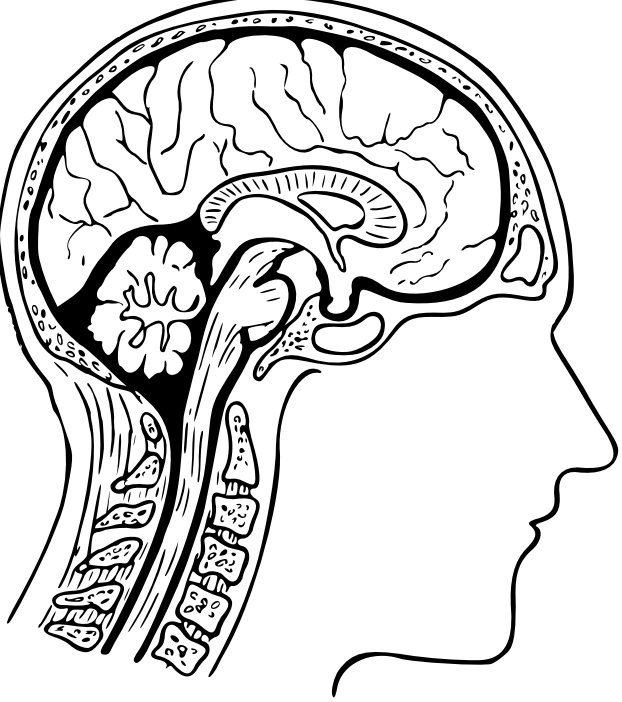
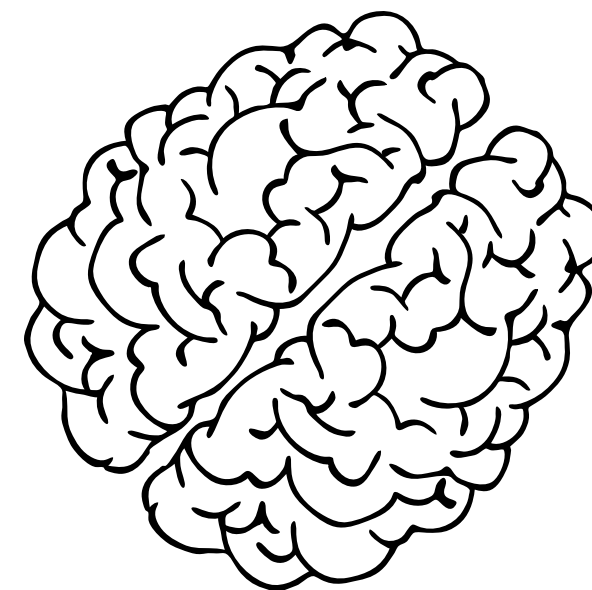
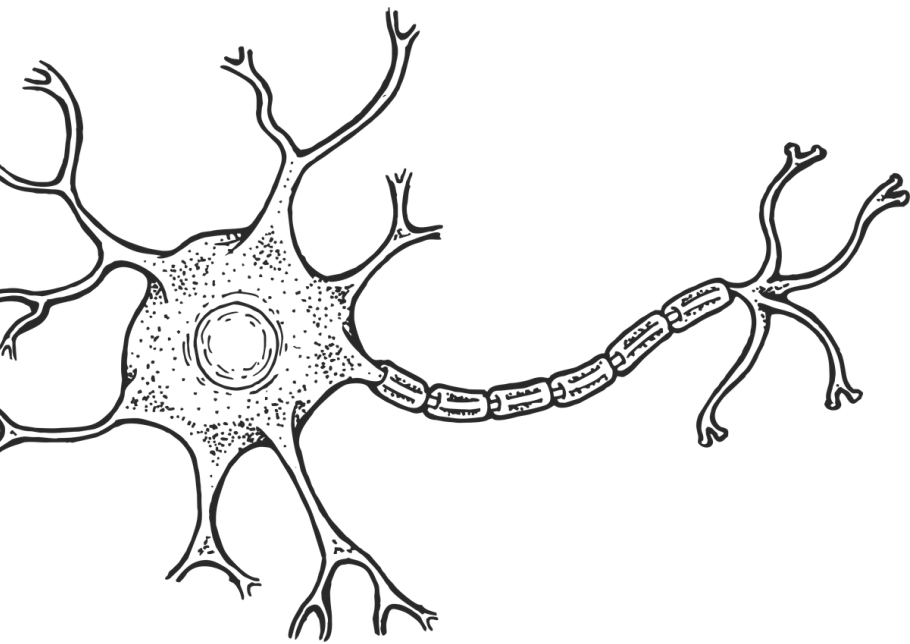


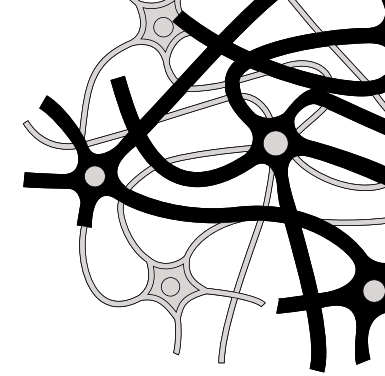
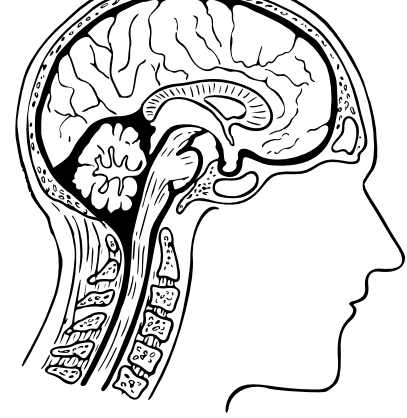


Session 21: Spectral Analysis

PSD and Welch's Method

presented by Lynn





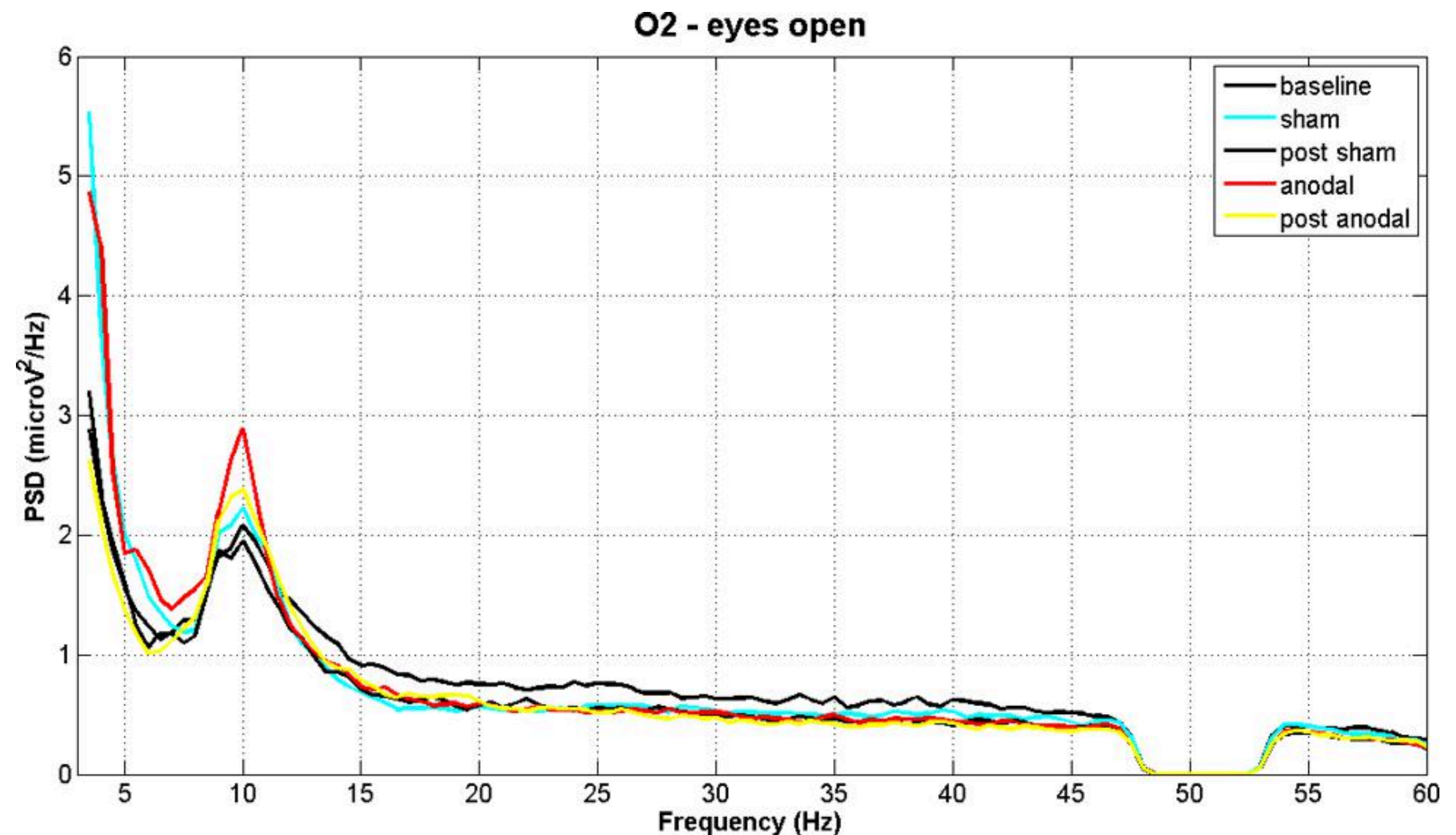
Session Objectives

Compute Spectral Density

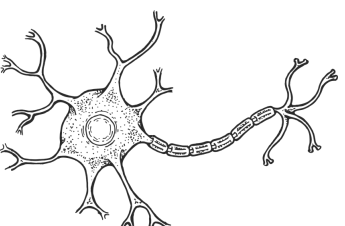
- Analyze the frequency content of EEG signals

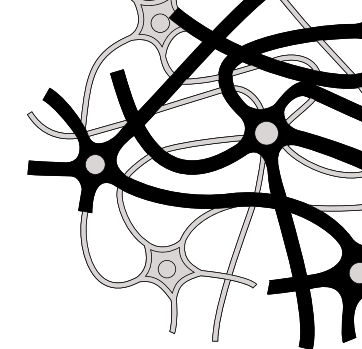
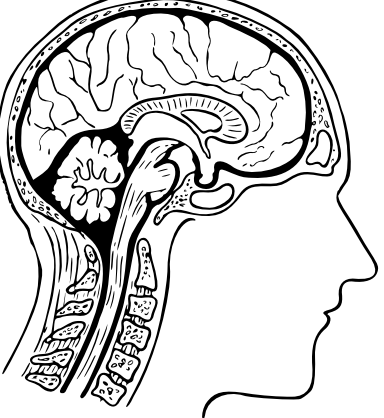
Quantify Band Power

- Identify features for psychiatric and neurological biomarkers.



Transitioning from time-domain to
frequency-domain analysis





Welch's Method in MNE

Averaging periodograms of overlapping windowed segments to reduce variance.

Core Concept

- Averages periodograms of overlapping segments. Reduces noise.

Window Size (n_fft)

- Long windows equal better frequency resolution, poorer temporal resolution.

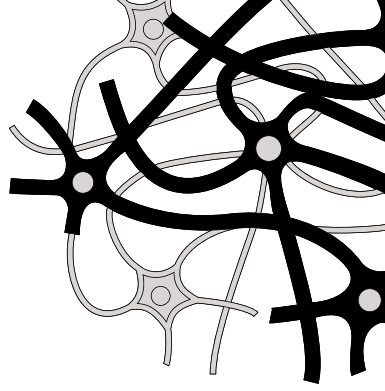
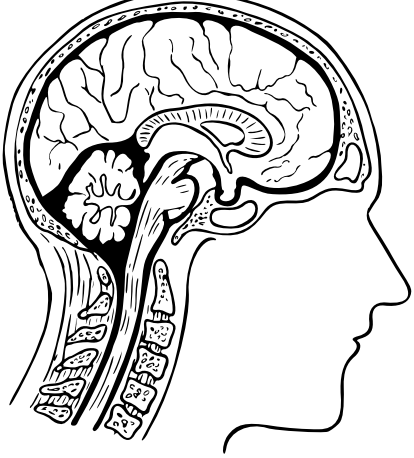
Overlap (n_overlap)

- Typically 50%. Prevents data loss at window edges from tapering.

MNE Implementation

```
epochs.compute_psd(method='welch')
```





Standard frequency bands

Delta (1-4Hz)

- Deep sleep, pathological states

Theta (4-8Hz)

- Drowsiness, cognitive control

Alpha (8-13Hz)

- Relaxed wakefulness.
- Prominent occipitally.

Beta (13-30Hz)

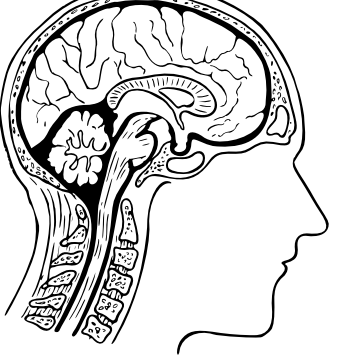
- Active thinking, motor execution

Gamma (30+ Hz)

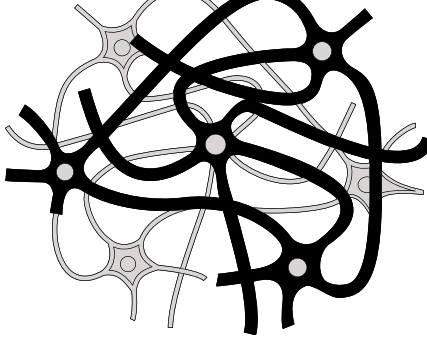
- High level cognitive processing

Extraction Method

- Define frequency boundaries, slice PSD array, and compute mean absolute power across specific channels.



Metric Export: MNE to Pandas



Translating multi-dimensional arrays(***Epochs*** x ***Channels*** x ***Frequencies***) to tabular data.

1

Bridge MNE and Pandas

- Translate sliced NumPy array into structured tabular format for downstream statistical testing.

2

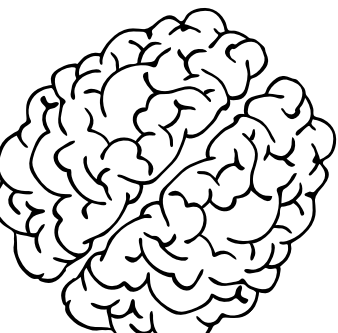
Map Conditions

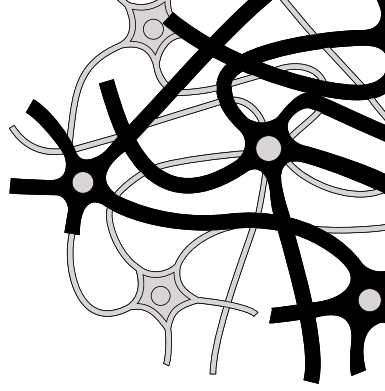
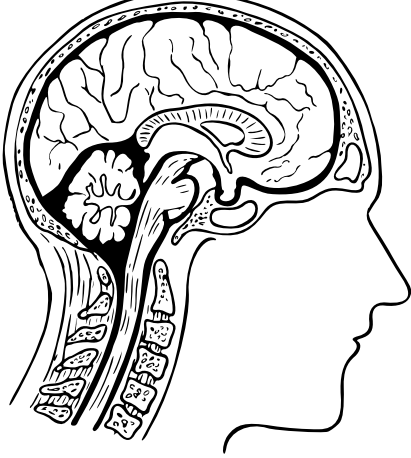
- Invert MNE event dict to map numeric event codes back to their descriptive string labels

3

DataFrame construction

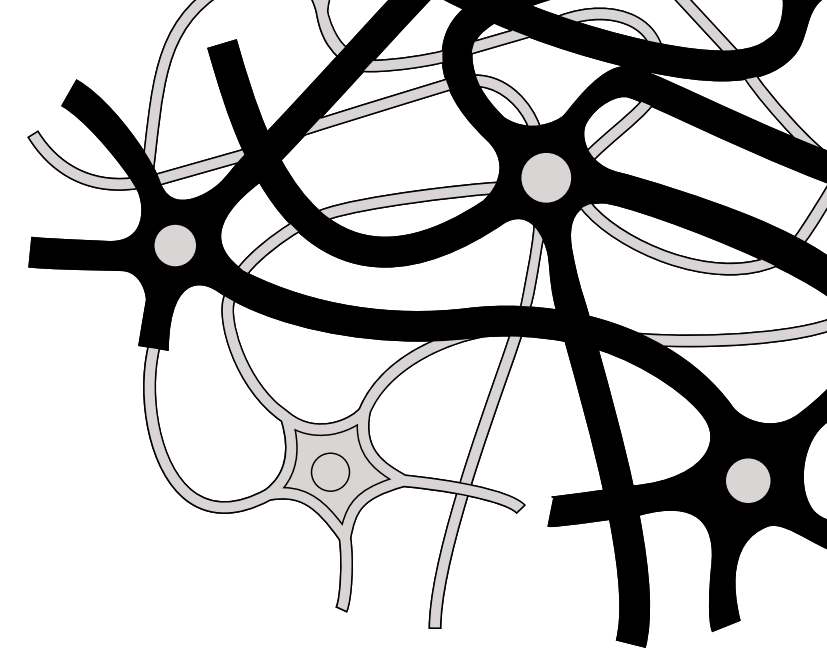
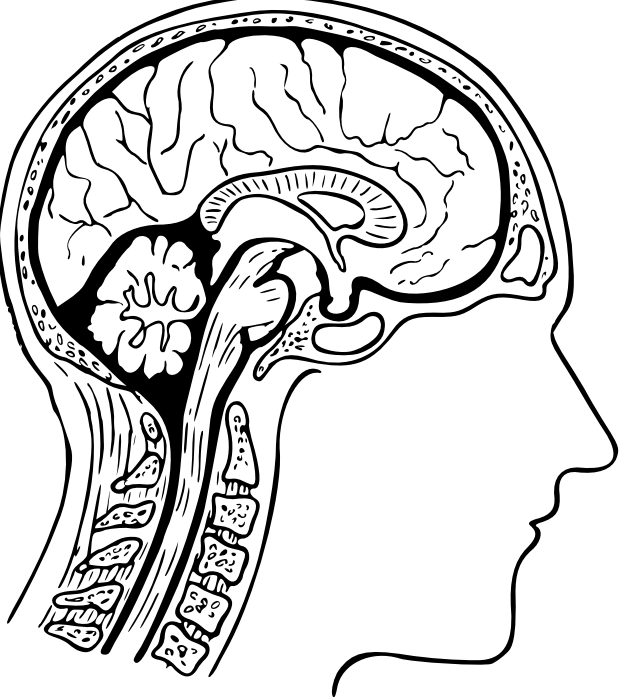
- Build df_spectral combining epoch ID, condition and extracted frontal_theta_power array.



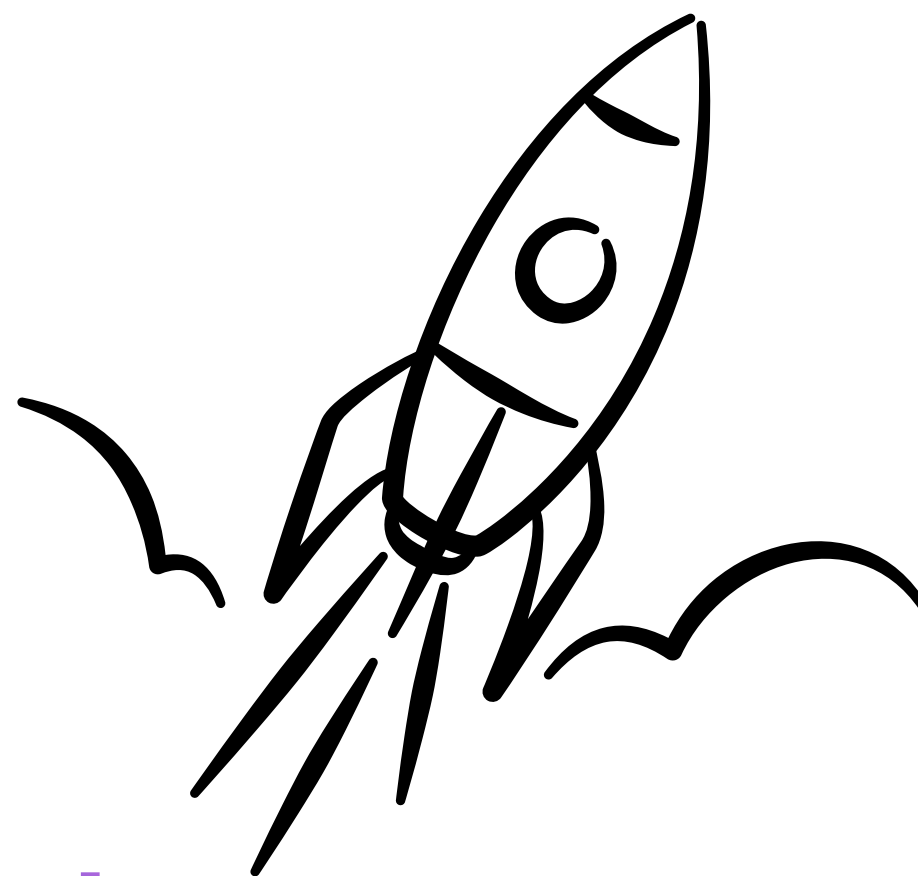
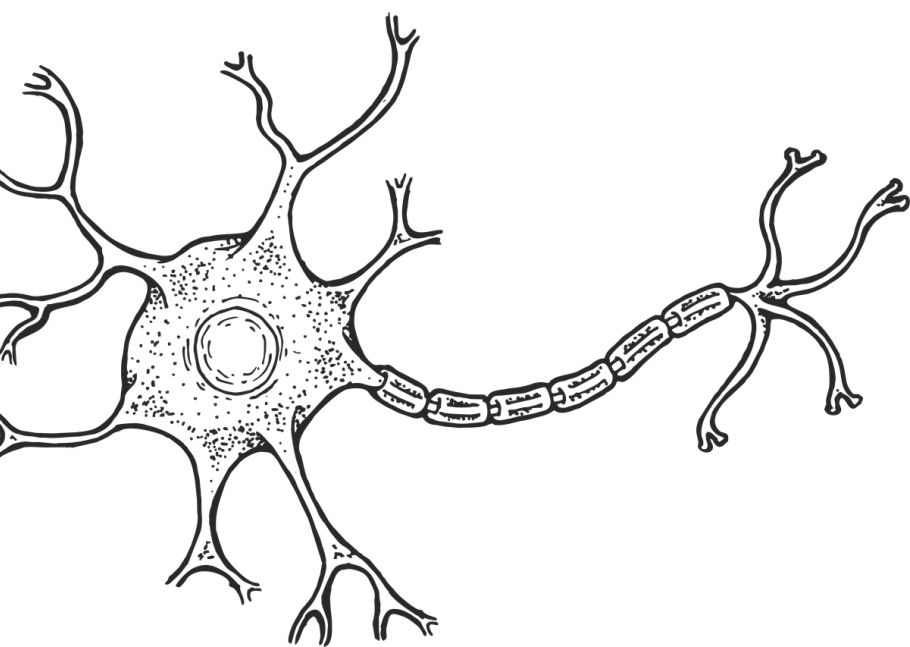


Class Activity

- **Task**
 - Replicate extraction workflow to isolate Alpha power(8-13Hz) over occipital channels.
- **Data Integration**
 - Add results to new column named occipital_alpha_power to existing df_spectral DataFrame
- **Verification**
 - Group updated Dataframe by condtion and print the mean Alpha power for Seizure vs Non seizure states
- **Output:**
 - Export finalised DataFrame to spectral_features.csv in reference to last session



To the Notebook!



Colab

